

COURTSIDE: Pricing the Value of Speed in In-Play Tennis Prediction Markets

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Abstract. After each point, a Polymarket tennis price (a win probability) is set by whoever learns the point first. In public data on 13,084 matches, wallets that repeatedly trade within 3 s of a point earn after fees in all 11 months, in and out of sample, while everyone else loses; copying them has no factor exposure and an out-of-sample Sharpe ratio of 6.7 that doubled fees erase. About half the edge is gone within 1–2 s, and the first second of video delay costs a simulated computer-vision (CV) trader \$42–74 a day; our pipeline needs 54 ms. With an assumed feed delay (no feed purchased), that trader makes, pre-registered, +\$15, +\$4 and −\$13 a day out of sample at 0.5, 1 and 3 s (+\$81, +\$57, −\$11 post hoc). Every blind test of a book we could trade failed.

JEL: G14, G12, C58. **Keywords:** prediction markets; latency arbitrage; in-play betting; tennis.

1 Summary

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- Each tennis point changes who is likely to win, and on Polymarket the price is that probability. Quotes lag the point by about a second, so whoever learns it first sets the price: the wallets that repeatedly trade in the first 3 s after a point (the *fast tier*, picked on past months) earn after fees in all 11 months, in and out of sample, and everyone else loses. Copying them has no factor exposure and an out-of-sample Sharpe ratio of 6.7, which doubled fees wipe out.
 - About half the edge is gone 1–2 s after the point (+1.18¢ a share in the block of the move, +0.59¢ at 1–2 s, in sample), and the first second of feed delay costs a simulated CV trader \$42–74 a day. Our pipeline adds 54 ms: with a 1 s feed an order can execute 2,119 ms after the point, inside the 3 s bar.
 - With an assumed feed delay, the pre-registered simulation makes +\$15, +\$4 and −\$13 a day out of sample at 0.5, 1 and 3 s, so it pays only with a feed near 1 s or faster (post hoc: +\$81, +\$57 and −\$11).
 - Every blind test of a book we could actually trade failed, and so did a replay on real order books; we report all 4,219 variants and 77 reads of held-out data, and used no real money.
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2 Economic hypothesis

Given how often each player wins a point on serve, a Markov model of points, games and sets gives the exact chance that each player wins from any score (Klaassen and Magnus, 2001). A Polymarket price is that chance in cents, so each point moves fair value by a known amount at a known moment. We call that amount the *swing* $\Delta(k)$ of point k : the chance of winning the match if point k is won, minus the chance if it is lost. It averages 5.7% and reaches 38.4% on a big point (Appendix A).

The news reaches traders in tiers: the court first, then video, then public score feeds 28.2–44.1 s after the price has moved (Fig. A3). A maker whose quote still rests after the point is picked off by

Rerun everything with bash reproduce.sh; rebuild this paper with python scripts/build_paper.py (<https://github.com/OjasMishra32/computervisionpredictivetabletennismoneymachineouuuushiii>). Paper trading only: no real money, no licensed feed, no footage of traded matches and no court camera.

whoever knows first: the adverse selection of Glosten and Milgrom (1985) and the latency race Budish et al. (2015) and Aquilina et al. (2022) measured in stocks. In-play betting prices react late (Croxxon and Reade, 2014; Brown, 2012). We add a wallet-level map of who is fast and a price for each second. Polymarket holds every *taker* order (one that hits a resting quote) for 1 s, so makers can cancel. The edge is shrinking: the fast tier grew from 4 to 131 wallets as its edge fell 0.20¢ a month ($t = -3.98$).

H6 (pre-registered). We expect wallets that trade within 3 s of a score move in Polymarket ATP, WTA and Challenger match markets to earn a positive net *markout* over 30 s (profit at the price 30 s later, after fees), because the other side is a maker who has not yet cancelled a stale quote. The edge persists because physics and data licences decide who learns first. If so, fast wallets picked on past months only stay positive every month and out of sample; H6 fails if the markout is ≤ 0 in over a third of months, or out of sample. H1–H4 were committed before any result (7232986, 2026-10-03 09:50 UTC); H6 was frozen before we opened the held-out data.

3 Data and universe

We use every resolved ATP, WTA and Challenger singles match market on Polymarket with at least \$5,000 traded, Oct 8, 2025 to Oct 3, 2026: 13,084 matches and \$2.84B of volume, from public order books and trade tapes (Polymarket, 2026a). *In sample* (IS) is the first 80% of matches by start time; *out of sample* (OOS) is the last 2,617, from 2026-08-25 14:15 UTC (the last 10.7% by time). We looked at the OOS while building v2 and the CV simulation, so it is *burned*. For a blind test, which also checks survivorship, we held back 11,307 smaller markets we never examined and opened them once per frozen rule. Six tapes hit the API’s offset cap, 2,889 early matches paid no fee, 2.89% of markets settled 50/50 (kept), tape timestamps run 1.98 s late, and no corporate actions apply. Timing data: ESPN and the WTA (ESPN, 2026; WTA, 2026); vision: OpenTTGames footage (Voeikov et al., 2020), BlurBall, WASB, TrackNet and court-keypoint weights (Gossard et al., 2026; Tarashima et al., 2023; Huang et al., 2019; yastrebkvs, 2026) and one Pexels tennis rally (Gelato Prod, 2021).

4 Methodology

The fast tier and v2. We flag a score move when the 10 s volume-weighted price moves at least 4¢ away from its prior 60 s. Each month a wallet joins the fast tier if, in earlier months, it traded at least 30 times 0–3 s after a flag, in at least 10 matches, with $t > 3$. Our copy book, v2, copies their trades, sizes each by its resolution risk (the best of 55 sizing rules, picked walk-forward), caps orders at \$1,000 and exposure at 100 shares net per match, and holds to resolution. At their own prices it measures what their speed is worth, not what we could earn: copying the same trades 3 s later loses (Fig. 2a).

Being fast ourselves. Our second strategy uses computer vision (CV): a model watches the video, calls the point as early as it can and sends a taker order before the stale quote is cancelled. We have no video of the matches we trade, so we simulate this trader on past points with an assumed feed latency (licensed feed not purchased); all other parameters are measured. The order wins only if

$$V + \delta + D - \ell < L + R \iff V < L + R - D - \delta + \ell, \quad (1)$$

where V is the feed delay (how late the video reaches us), δ our pipeline plus network time, D the 1 s taker hold, ℓ how early the CV calls the point, L the *stamp lag* (how long after the ball lands the umpire’s official point stamp is logged) and R when the book reprices relative to that stamp. On 482 points the book moved 1.2 s before the stamp. Nobody publishes L , so we show two readings: 2.0 s, which we *pre-registered* (wrote down before seeing any result; PREREG.md, 17:25 UTC), and 3.14 s, a *post hoc* estimate from one day of data. The simulated trades are past points where the price later moved at least 4¢, so they are *selected on outcomes, not ex ante*.

Our CV is real on table tennis. On 41 held-out balls that went out (120 fps), our live causal engine called 4 early (median lead 162 ms) and none wrongly, at 119.9 fps on one L4 GPU. On a freely licensed real tennis rally (Gelato Prod, 2021), our tennis tracker found the ball in 89% of 300 frames (48/48

Table 1. *Our copy of the fast tier's trades (v2), at their prices: the edge holds out of sample but is thin.*

Return	IS	OOS	Risk and costs	IS	OOS
Net ¢ a share	+1.38	+0.60	Max drawdown	-2.0%	-2.1%
95% CI	[1.17, 1.59]	[0.09, 1.13]	Worst month	+\$2,543	-\$434
Sharpe ratio	14.5	6.7	Skew	+0.53	+0.33
95% CI	[11.9, 17.4]	[1.9, 12.3]	Turnover, × a year	93	129
Deflated Sharpe	0.997	0.075	Months positive	7/7	2/3
Annual return	253%	148%	Net ¢, fees ×2	+0.77	-0.34
Annual volatility	17.5%	22.2%	Net ¢, all costs ×2	+0.27	-0.84

Notes: 55,662 trades in 206 days IS (+\$40,426), 10,412 in 40 OOS (+\$3,688; burned, not blind), net of costs. Sharpe: daily P&L $\times \sqrt{365}$; CIs: block bootstrap; deflated at 3,386 trials; formulas: Appendix A.

random spot-checks on the ball in play) and fitted the court lines in every frame, a median 0.36 px off the paint; we make no in/out calls on single-camera tennis footage. On our own footage a frame becomes a built order in 54 ms, and with a simulated 1 s feed, 65 ms of network and the 1 s hold the order can execute 2,119 ms after the point, inside the organizers' 3 s bar (Fig. 1).

Costs and validation. Polymarket charges takers $f(q) = r q (1 - q)$ a share at price q , which is 10,000 $r (1 - q)$ bps of notional. Today $r = 5\%$: a trade at 50¢ pays 1.25¢ a share. Each match pays its own fee, every trade the spread it crossed; we also double both. v2 paid 120 and 179 bps in fees (IS, OOS) and still netted 273 and 114 bps. We tune only in sample, walk-forward (fit on past months, test on the next).

5 Results

The copy book. The fast tier earns every month and everyone else loses (Fig. 2a). v2 earns in both periods (Table 1), but thinly: entering half a tick worse leaves +0.10¢ a share out of sample, and doubling fees turns it to -0.34¢ (Fig. 2b). It stays positive in every fee and hold regime, from +1.96¢ (3 s hold, no fee) to +1.02¢ (1 s hold, 5% fee; Fig. A1).

A Sharpe ratio of 14.5 is a red flag. We assumed a bug and found one: the detector's start time used hindsight (deviation D9), and fixing it cut the in-sample Sharpe from 16.8 to 14.5. The rest comes from about 270 small, nearly independent bets a day (mean correlation within a day 0.0013). The *deflated Sharpe* (Bailey and López de Prado, 2014), the chance the Sharpe is real once every variant is counted, is 0.997 in sample but only 0.075 out of sample: forty days cannot rule out luck. Nearby settings work too: all 55 sizing rules we tried have a per-share 95% CI above zero in sample (Sharpe 4.1–16.8), the v2-safe grid's probability of backtest overfitting is 9–24% across block choices (A6), and the CV profit falls across all 14 feed delays from 0 to 60 s, never rising by more than \$0.09 a day (Fig. 3).

Frame to executable order in 2.12 s; our part 54 ms

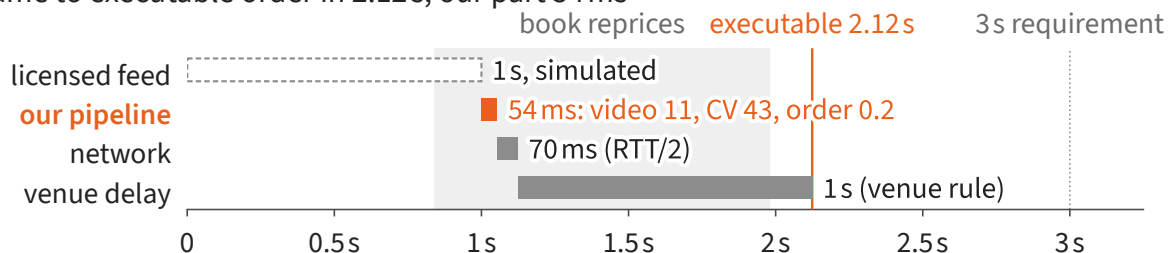


Figure 1. *We can trade inside 3 s. Mean stage times over 24 runs on our own footage; the 1 s feed is simulated; order built, not sent. Shaded: when the book reprices. Source: results/e2e.*

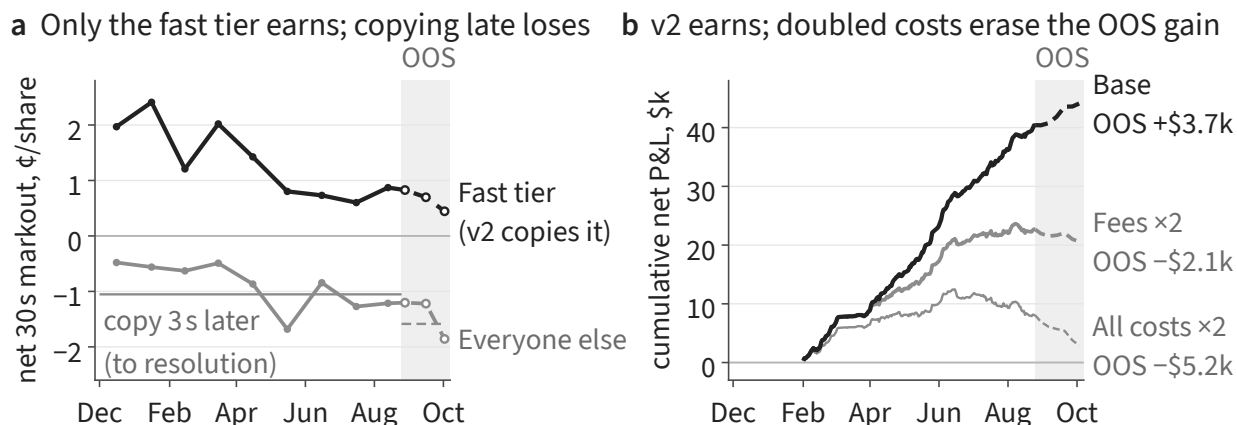


Figure 2. Only the fast tier earns, and the edge is thin. (a) Monthly net 30 s markout (copying 3 s later: held to resolution). (b) v2's net P&L, base and doubled costs. Source: results/v2.

Table 2. The CV strategy at three assumed feed delays: pre-registered it breaks even near 1 s.

Feed delay V	In sample			Out of sample (burned)		
	\$ a day	Sharpe	¢ a share [95% CI]	\$ a day	Sharpe	¢ a share [95% CI]
A. Pre-registered stamp lag, $L = 2.0$ s; break-even V: 1.09 s IS, 1.01 s OOS						
0.5 s, best case	+\$28	4.0	+0.61 [0.08, 1.14]	+\$15	2.1	-0.02 [-1.22, 1.09]
1 s, base case	+\$15	2.1	+0.40 [-0.32, 1.08]	+\$4	0.3	-0.38 [-2.08, 1.21]
3 s, the organizers' bar	-\$13	-2.6	-1.35 [-3.15, 0.37]	-\$13	-2.6	-1.69 [-4.85, 1.32]
B. Post hoc estimate, $L = 3.14$ s (95% CI 2.23–3.22 s); break-even V: 2.23 s IS, 2.14 s OOS						
0.5 s, best case	+\$133	15.5	+1.21 [1.00, 1.42]	+\$81	12.2	+0.82 [0.39, 1.25]
1 s, base case	+\$94	11.9	+1.11 [0.85, 1.38]	+\$57	8.8	+0.66 [0.10, 1.19]
3 s, the organizers' bar	-\$6	-1.4	-0.79 [-2.26, 0.60]	-\$11	-2.4	-1.47 [-4.45, 1.34]

Notes: simulated, assumed feed latency (licensed feed not purchased); parameters measured; 20 seeds a cell. Trades: past points where the price later moved $\geq 4\text{¢}$, selected on outcomes, not ex ante. Call rates are the offline rule's (look-ahead feature); the live engine's own calls give +\$4 and +\$57 OOS at 1 s. Per-tournament timing came after the first P&L (deviation V3).

What speed is worth. Pre-registered, the CV trader breaks even at a 1.01 s feed delay out of sample (Table 2, Fig. 3): the call must reach the venue 0.89 s before the umpire's stamp. Post hoc it makes +\$57 a day at 1 s (+\$8 at the low end of the stamp-lag interval), but read per point the post hoc inference loses (-\$17 a day). Every reading loses at 3 s, so meeting the 3 s bar is not enough. A replay of 9 matches recorded live on Oct 3 against their real books, calling every point ex ante, loses in 36 of 36 settings (-0.94¢ a share at 1 s; Fig. A2): the gap is knowing which points matter in advance.

What failed. Every blind test of a book we could trade failed: v2 on the 11,307 unseen markets (+1.22¢ [-0.19, 2.65]), the pre-registered maker book and the CV rule v3 (-0.52¢); Appendix B lists every test. We tried 4,219 variants, read held-out data 77 times, changed rules after a look 2 times for v2 and 13 for the CV simulation, and chose the post hoc reading and 1 s base case after looking. Blind forward test: pre-registered but not run within the hackathon window (HYPOTHESIS_V2.md A5).

6 Risk management

Limits and de-risking. Each order is at most \$1,000 and each match at most 100 shares net, at prices from 5¢ to 95¢, so one match can lose at most \$95 plus fees; a \$1,000 daily loss ends the day,

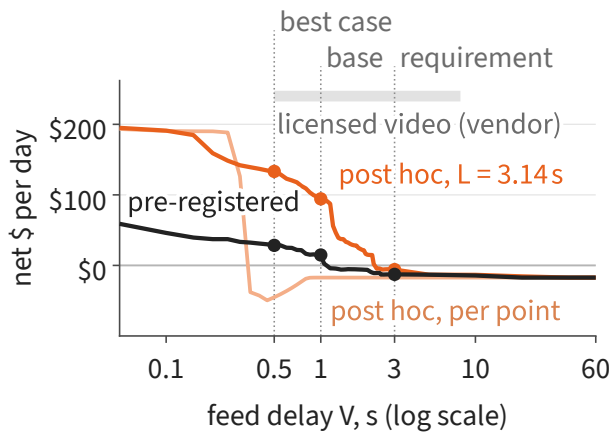
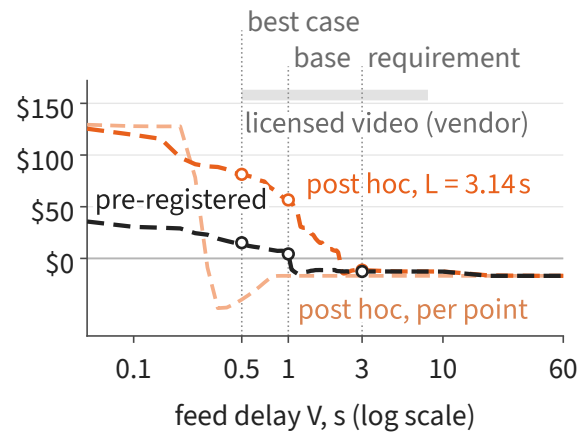
a IS: zero profit at 1.1–2.2 s of delay**b** OOS: zero profit at 1.0–2.1 s of delay

Figure 3. Each second of feed delay costs money. Simulated CV strategy, (a) IS, (b) burned OOS; dots: Table 2; grey bar: vendor-stated licensed video. Assumed feed latency (licensed feed not purchased); parameters measured. Source: results/tier0.

all in code. Set in advance, we stop if the score feed is over 2 s stale, vision over 1 s stale, or latency above its rolling 95th percentile; we halve size if the trailing 30-day edge falls below 0.3¢, and stop if it reaches zero, if drawdown hits 5%, or if the venue changes its fee or hold.

Model, concentration and tail risk. Wrong CV calls always fill, so an out call trades only at confidence ≥ 0.95 and one confirmed false call halts the day. Phantom calls outside play come about 30 an hour; 2.5 a day would erase the pre-registered 1 s OOS P&L (32 post hoc). Replayed on the engine's held-out call log, our rally gate (trade a miss call only within 2.0 s of a bounce call, set before the test) removes 5 of the 7 calls on unlabelled balls (4 of the 5 between rallies) but keeps only 3 of 5 correct calls (2 at 0.5–0.8 s), leaving about 8 phantom calls an hour: not yet safe to trade. The top five copied wallets carry 81% of v2's P&L in sample and 142% out of sample. The UMA oracle settles markets (Polymarket, 2026c), US persons may only close positions (Polymarket, 2026b), and courtsiding breaches ticket terms (International Tennis Federation, 2026). Factor risk is nil (largest factor $|t| = 1.36$; α has $t = 8.48$) (Fama and French, 1993; Carhart, 1997).

7 Liquidity and capacity

v2 trades \$7,182 a day, 0.07% of match volume; its OOS confidence interval stays above zero only up to $1\times$ size (\$22,754 of capital), and $5\times$ loses. The CV book has no capacity at the pre-registered lag (Sharpe 2.7 IS, 0.8 OOS even at its smallest size). Post hoc, taking at most half the stale depth at each point, its Sharpe halves at \$40,000 of capital OOS and \$73,000 IS and peaks at \$177 a day on \$83,000 OOS (Fig. A4). What binds is the few seconds after each point: the book is 0.22% of in-play volume but 9% of what the fast tier trades 0–3 s after a point. Data is the other limit: at 1 s the book could pay at most \$1,644 a month for data post hoc and \$55 pre-registered, against reported feed prices of \$1,250–\$10,000. At today's fee COURTSIDE prices speed; it is not yet a business.

8 Limitations and next steps

The stamp lag L is unmeasured and moves the break-even one for one (Eq. 1); the 1 s hold dominates it, since even for a 20 ms bot the inferred lag is 2.85 s. One session with a licensed feed and a few test orders would measure L ; at 1 s the feed pays for itself only if $L \geq 3.0$ s, so buying it is a business decision we have not made. Our CV call accuracy comes from table tennis, the trading is simulated and we never sent an order. A fee rise, a shorter hold or more fast wallets would break the strategy.

References

- Ably (n.d.). Case study: Genius Sports. <https://ably.com/case-studies/genius-sports>.
- Aquilina, M., Budish, E., and O'Neill, P. (2022). Quantifying the high-frequency trading “arms race”. *Quarterly Journal of Economics*, 137(1):493–564.
- Bailey, D. H., Borwein, J., López de Prado, M., and Zhu, Q. J. (2017). The probability of backtest overfitting. *Journal of Computational Finance*, 20(4):39–69.
- Bailey, D. H. and López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107.
- Brown, A. (2012). Evidence of in-play insider trading on a UK betting exchange. *Applied Economics*, 44(9). Cited from its abstract.
- Brown, A. (2014). Information processing constraints and asset mispricing. *Economic Journal*, 124(575).
- Budish, E., Cramton, P., and Shim, J. (2015). The high-frequency trading arms race: Frequent batch auctions as a market design response. *Quarterly Journal of Economics*, 130(4):1547–1621.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52(1):57–82.
- Croxson, K. and Reade, J. J. (2014). Information and efficiency: Goal arrival in soccer betting. *Economic Journal*, 124(575):62–91.
- ESPN (2026). Public tennis scoreboard endpoint. Read live on 2026-10-03 for latency timing only; derived timings committed.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- French, K. R. (2026). Data library: Fama–French factors and momentum. https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.
- Gaming Intelligence (2026). Genius sports agrees Polymarket prediction market deal. <https://www.gamingintelligence.com/finance/234424-genius-sports-agrees-polymarket-prediction-market-deal/>. 5 August 2026; Polymarket US, selected leagues.
- Gelato Prod (2021). Tennis players playing match (video 10378830). Pexels, <https://www.pexels.com/video/tennis-players-playing-match-10378830/>. Pexels License; downloaded October 3, 2026 and trimmed to a 10 s rally.
- GL Systemhaus (2018). Who cheers first? About latencies in sports livestreaming. <https://www.gl-systemhaus.de/en/blog/who-cheers-first-about-latencies-in-sports-livestreaming>.
- Glosten, L. R. and Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1):71–100.
- Gossard, T., Radovic, F., Ziegler, A., and Zell, A. (2026). BlurBall: Joint ball and motion blur estimation for table tennis ball tracking. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. arXiv:2509.18387; code and weights MIT.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1):5–68.
- Harvey, C. R., Rattray, S., Sinclair, A., and Van Hemert, O. (2017). Man vs. machine: Comparing discretionary and systematic hedge fund performance. *Journal of Portfolio Management*, 43(4):55–69.
- Huang, Y.-C., Liao, I.-N., Chen, C.-H., Ik, T.-U., and Peng, W.-C. (2019). TrackNet: A deep learning network for tracking high-speed and tiny objects in sports applications. In *IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*.
- International Tennis Federation (2026). ITF world tennis tour regulations 2026.

- International Tennis Integrity Agency (2026). Tennis anti-corruption program 2026.
- ISPreview (2021). Broadcast lag in live online TV sport streaming frustrates fans. <https://www.ispreview.co.uk/index.php/2021/06/broadcast-lag-in-live-online-tv-sport-streaming-frustrates-fans.html>.
- Kalshi (2026). Public market-data API. <https://trading-api.readme.io>. Accessed October 2026.
- Klaassen, F. J. G. M. and Magnus, J. R. (2001). Are points in tennis independent and identically distributed? Evidence from a dynamic binary panel data model. *Journal of the American Statistical Association*, 96(454):500–509.
- Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.
- NEXT.io (n.d.). Matt fleckenstein: raising the bar in in-play betting (Genius BetVision). <https://next.io/news/betting/matt-fleckenstein-raising-bar-in-play-betting/>. Vendor statement, 4–8 s venue to phone; unverified.
- Politis, D. N. and Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313.
- Polymarket (2026a). Gamma, data and CLOB APIs, sports websocket, fee and rate-limit documentation. <https://docs.polymarket.com>. Public, no keys; accessed October 2026.
- Polymarket (2026b). Geographic restrictions (close-only jurisdictions). <https://docs.polymarket.com/api-reference/geoblock>. Accessed October 2026; docs/RISK.md R16.
- Polymarket (2026c). Resolution: the UMA optimistic oracle; pUSD collateral. <https://docs.polymarket.com/developers/resolution/UMA>. Accessed October 2026; docs/RISK.md R12–R13.
- PR Newswire (2026). Polymarket secures exclusive ATP Tour streaming rights for prediction markets. <https://www.prnewswire.com/news-releases/polymarket-secures-exclusive-atp-tour-streaming-rights-for-prediction-markets-302841534.html>. 3 August 2026.
- Regen Sports (2026). I attended Sportradar’s game, set, tech. <https://regensports.substack.com/p/i-attended-sportradars-game-set-tech>.
- Sportradar (n.d.). Live streams for betting operators. <https://sportradar.com/betting-gaming/products/live-streams/>. Vendor page.
- Stats Perform (n.d.). Realtime streaming; official WTA data for sportsbooks. <https://www.statsperform.com/products/realtime-streaming/>. Vendor claim, unverified.
- Tarashima, S. et al. (2023). Widely applicable strong baseline for sports ball detection and tracking. In *British Machine Vision Conference (BMVC)*. WASB-SBDT, MIT.
- The Desk (2026). Stats perform and phenix measure latency at Super Bowl LX. <https://thedesk.net/2026/02/stats-perform-phenix-latency-super-bowl-lx/>.
- Voeikov, R., Falaleev, N., and Baikulov, R. (2020). TTNNet: Real-time temporal and spatial video analysis of table tennis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. OpenTTGames dataset, CC BY-NC-SA 4.0.
- WTA (2026). Public scores and point-by-point API. Read live on 2026-10-03 for latency timing only; derived timings committed.
- yastrebkv (2026). TrackNet, TennisProject and TennisCourtDetector: tennis ball and court-keypoint models. <https://github.com/yastrebkv/TennisProject>. Commits 730ea17, b7552e9 and e5cd4f1, accessed October 2026; no licence file, weights and code used unmodified for evaluation only.
- Disclosure.** We used no real money and connected no funded account. Simulated parts are labelled where they appear. We hold no licensed data feed, no footage of the matches we trade and no court access. Open-source parts we did not build: the OpenTTGames dataset, the BlurBall and WASB model

weights, and the TrackNet weights and TennisProject/TennisCourtDetector code (used unmodified, for evaluation only), cited above. The tennis rally is a Pexels stock clip (Gelato Prod, 2021); the logo on page 1 is the Gator Quant Hacks 2026 event's own mark. Authors: Ojasva Mishra, Yoan Exposito, Rafael Penhas, Ian Hoang (University of Florida).

Appendix A Calculations

Everything needed for scoring is in the main pages. Below: formulas with worked examples (A), every test (B), extra figures (C) and the records (D). Each value is in results/paper/numbers.json with its source file.

A1. Sharpe ratio. $SR = \sqrt{365} \bar{d} / s_d$, where d_t is net P&L on every calendar day (zero on days without trades) and s_d its standard deviation. v2 IS: $\bar{d} = \$196.24$, $s_d = \$258.87$ over $T = 206$ days, so the daily ratio is 0.758 and $SR = 0.758 \times 19.105 = 14.48$. OOS: $\$92.19 / \$264.15 \times 19.105 = 6.67$.

A2. Sortino ratio. $\sqrt{365} \bar{d} / DD$, where the downside deviation is $DD = \sqrt{\frac{1}{T} \sum_t \min(d_t, 0)^2}$. For v2 OOS, $DD = \$125.64$, so the ratio is $\sqrt{365} \times \$92.19 / \$125.64 = 14.0$.

A3. Maximum drawdown. $MDD = \min_t (E_t - \max_{s \leq t} E_s)$, where E_t is cumulative P&L, as a share of capital. v2 OOS: $-\$469 / \$22,754 = -2.06\%$.

A4. Probabilistic Sharpe. The chance that the true daily Sharpe beats SR^* , given the sample length T , skew γ_3 and kurtosis γ_4 (Bailey and López de Prado, 2014):

$$PSR(SR^*) = \Phi\left((\hat{SR} - SR^*)\sqrt{T-1} / \sqrt{1 - \gamma_3 \hat{SR} + \frac{\gamma_4-1}{4} \hat{SR}^2}\right).$$

v2 OOS: $\hat{SR} = 0.349$, $T = 40$, $\gamma_3 = 0.326$, $\gamma_4 = 3.26$, so $PSR(0) = 0.987$.

A5. Deflated Sharpe. $DSR = PSR(SR_0)$, where SR_0 is the best Sharpe we would expect from N tries on pure noise, with γ the Euler-Mascheroni constant:

$$SR_0 = \sqrt{V} \left((1 - \gamma) \Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma \Phi^{-1}\left(1 - \frac{1}{N_e}\right) \right).$$

v2 OOS with $N = 3,386$ and $V = 1/(T-1)$: the two Φ^{-1} terms are 3.436 and 3.698, $SR_0 = 0.574$ a day (10.97 a year), and $DSR = 0.075$. Table 1 shows the lowest DSR over three choices of V .

A6. Probability of backtest overfitting. (Bailey et al., 2017). Split the daily P&L of 24 variants into $S = 16$ blocks. For each of the $\binom{16}{8} = 12,870$ ways to call half the blocks in sample, take the best in-sample variant, find its rank ω out of sample and set $\lambda = \ln \frac{\omega}{1-\omega}$; $PBO = \Pr(\lambda \leq 0)$. v2-safe grid: $PBO = 15\%$ (median $\lambda = 1.27$).

A7. Stationary block bootstrap. (Politis and Romano, 1994). Resample days in blocks of random length (mean 5 days) 10,000 times and read the percentiles. v2 OOS Sharpe: [1.9, 12.3], standard error 2.65, $P(SR \leq 0) = 0.0023$.

A8. Factor regression. $r_t = \alpha + \beta_M \text{MktRF}_t + \beta_S \text{SMB}_t + \beta_H \text{HML}_t + \beta_U \text{Mom}_t + \varepsilon_t$ on daily return on capital, with Newey-West errors (5 lags) (Fama and French, 1993; Carhart, 1997; Newey and West, 1987; French, 2026). v2 IS (138 days): $\alpha = 0.801\%$ a day ($t = 8.48$); $\beta_M = 0.173$ ($t = 1.36$), $\beta_S = 0.085$, $\beta_H = 0.044$, $\beta_U = 0.028$; $R^2 = 3.3\%$.

A9. Taker fee. $f(q) = r q(1-q)$ a share at price q , or $r(1-q)$ per dollar, which is 10,000 $r(1-q)$ bps. At $r = 5\%$: $q = 0.5$ pays 1.25¢ (250 bps) and $q = 0.8$ pays 0.80¢ (100 bps).

A10. Point swing. $\Delta(k)$ of Section 2, with the match-win probability computed exactly by recursion over points, games, tiebreaks and sets (src/markov.py). Two equal players who win 0.645 of service points: at the first point $\Pr(\text{win}) = 50.0\%$, moving to 51.2% or 47.8%, so $\Delta = 3.4\%$. Serving at 5-5, 30-40 in the deciding set, $\Pr(\text{win}) = 33.7\%$, moving to 47.3% or 8.9%, so $\Delta = 38.4\%$. The mean $|\Delta|$ over a match is 5.7%.

A11. Latency budget. Eq. 1. With the pre-registered $L = 2.0$ s, $D = 1$ s and our measured $\delta = 0.119$ s, a point whose book reprices at the median $R = -1.16$ s needs $V < -0.28 + \ell$: no feed wins it without an early call. At the 90th percentile, $R = 0.08$ s and $V < 0.96$ s. The sweep's break-even (1.09 s IS) comes from these late reprices.

A12. Capital and return on capital. Capital is three times the peak dollars locked in open positions, each held 4 h or until resolution. v2 IS: peak \$9,434, capital \$28,302; return = $365 \bar{d} / \text{capital} = 365 \times \$196.24 / \$28,302 = 253\%$ a year.

A13. Capacity and volume share. Capacity is the capital at which the seed-mean Sharpe falls to half its smallest-size value; share is our notional a day over market volume a day. CV, post hoc, IS, just below half Sharpe: \$10,428 a day against \$4.71M of in-play volume (0.22%) and \$117,047 traded by the fast tier 0–3 s after a point (9%).

Appendix B Everything we tested

Table A1. Every test we ran, its best result and our verdict. IS: in sample; OOS: out of sample (burned unless marked blind); ¢: net cents a share.

Test	What it was	Best result	Verdict
<i>Who is fast, and what they earn (public tapes)</i>			
H1 follow the jump	take the direction of a ≥ 4 ¢ move, hold 30 s	OOS -2.08 ¢ $[-2.17, -1.98]$; 20 of 20 IS variants below zero	rejected
H2 favourites	buy in-play favourites priced 85–97¢	OOS $+0.80$ ¢ $[-1.36, 2.66]$	no edge
H4 score feeds	do public score feeds lag the book?	median lag 44.5 s ($n = 75$)	confirmed
H5 quote after jumps	post maker quotes after a move	OOS $+0.16$ ¢ $[-0.49, 0.87]$	not significant
H6 fast tier	wallets trading 0–3 s after a move, picked on past months	$+0.93$ / $+0.75$ ¢ net 30 s markout (IS / OOS); 9/9 and 3/3 months positive	confirmed
Kalshi lead-lag	Kalshi vs Polymarket on the same points	Kalshi first in 68.9% of 106,347 repricings (median 5 s)	lag mostly mechanical
Signal decay (TT5)	with-jump markout at 1–2 s minus 10–30 s after a move	OOS $+0.66$ $[0.45, 0.87]$ ¢; whole year $+0.19$ $[-0.26, 0.46]$ ¢; fast tier $+1.18$ to $+0.59$ ¢ (IS, 0 to 1–2 s)	decays in today's regime only
Clocks	when the book, the stamp and the tape move	book 1.2 s before the stamp ($n = 482$); tape 1.98 s late	measured
Factor regression	Fama–French three factors and momentum	$\alpha t = 8.48$; largest factor $ t = 1.36$	no factor exposure
<i>Copying the fast tier (v2 and relatives)</i>			
v1	copy the fast tier, dollar sizing; OOS opened once	IS $+1.16$ ¢; OOS $+0.57$ ¢, $-\$36,056$	failed
Sizing	55 sizing policies, walk-forward	G_50pct_net100: Sharpe 14.5 IS; all 55 with IS CI above zero (Sharpe 4.1–16.8)	adopted (v2); plateau
Exits	maker exit instead of holding (902 variants)	IS $+0.53$ ¢ $[0.42, 0.64]$ (1 s / 5% regime); live fills adverse (27% filled in 30 s)	not adopted

Test	What it was	Best result	Verdict
Trade selection	which fast-tier trades to take (341 variants)	nested walk-forward +1.76¢ [1.36, 2.16] vs +1.16¢ (IS)	IS only
Side markets	side markets vs the match market (68 variants)	leaning maker +2.73¢ [1.73, 3.70] (IS); taking stale side quotes: no edge	IS only; tiny
v2	frozen copy book at the fast tier's own prices	Sharpe 14.5 / 6.7; OOS +0.60¢ [0.09, 1.13]	headline (Table 1)
v2, worse entry	enter ½ or 1 tick worse	IS +0.88 / +0.38¢; OOS +0.10 / -0.40¢	about zero OOS
v2, costs doubled	fees ×2 / all costs ×2	IS +0.77 / +0.27¢; OOS -0.34 / -0.84¢	fails OOS
v2, fixed costs	after central data and running costs	IS +\$29 a day; OOS -\$75 a day	fails OOS
v2, unseen markets	11,307 markets we never examined (blind)	IS +2.02¢; OOS +1.22¢ [-0.19, 2.65]	fail
v2-safe	net cap 50, fixed after v2's OOS losses	Sharpe 15.7 / 9.3; blind +1.21¢ [-0.02, 2.43]	fail (blind)
Maker v1	pre-registered maker book (blind)	+1.87¢ [-0.14, 3.86] per fill, -\$379	fail
Rigor pack	deflated Sharpe, PBO, block bootstrap	DSR 0.997 / 0.075; PBO 15% (9–24% by block choice); $P(SR \leq 0) = 0.0023$	luck not ruled out OOS
Blind forward test	v2 on new matches, CV rule v3 as secondary	pre-registered but not run within the hackathon window (HYPOTHESIS_V2.md A5)	not run
<i>Being fast ourselves (computer vision)</i>			
H3 early call	table-tennis MISS call 50 ms before contact, precision ≥ 0.95	live causal engine: 4 of 41 misses, none wrong; offline evaluation (look-ahead feature): 11/11	precision passes; recall low
Spin model, table tennis	spin-aware early call, held-out footage	10/12 right at 50 ms vs the frozen model's 11/11 (both offline)	not adopted
Tennis tracker, real clip	our tennis tracker on a Pexels rally (no labels)	ball in 89% of 300 frames; 48/48 spot-checks on the ball; court lines 0.36 px (median)	tracks; no in/out calls
Tennis spin model	spin-aware landing model (simulated physics)	0.57 vs 5.76 cm at 200 ms; OUT precision 1, recall 0.98	simulation only
Rally gate	a miss call trades only within 2.0 s of a bounce call (engine log; 0.5–1.2 s also run)	removes 5 of 7 calls on unlabelled balls; keeps 3 of 5 right ones (2 at 0.5–0.8 s)	not enough
GPU engine	streamed causal engine on one L4	119.9 fps; 4.6 ms p50; 0 frames dropped	measured
WebRTC pipeline	our clip over WebRTC into the engine	46 ms p50, 67 ms p99 frame to call; video lag 4–8 ms	measured
End to end	video frame to unsigned order on a live book	54 ms ours; 2,119 ms with a 1 s feed vs 3,000 ms	pass (paper)
Courtside camera	CV trader with a camera at the venue ($V = 0$)	+\$89 / +\$46 a day (pre-registered)	simulated; no camera

Test	What it was	Best result	Verdict
Latency sweep	V from 0 to 60 s, six stamp-lag readings (944 cells)	pre-registered break-even 1.01–1.09 s	Table 2
CV rule v3	rule tuned on 360 IS variants	IS -0.22¢ ; blind -0.52¢	fail
Replay, all points	9 matches' real books, every point called	-0.94¢ $[-1.28, -0.58]$ at $V = 1\text{ s}$; 36 of 36 settings below zero	loses
Replay, big points	only points with a large Markov swing (exploratory)	-0.95¢ $[-1.64, -0.40]$; 9 of 9 cells below zero	loses
Capacity	size grid for the CV book and v2	post hoc Sharpe halves at $\$40,000$ – $\$73,000$; pre-registered: none	small
Table-tennis markets	TT1–TT5 on Polymarket table tennis	no fast wallet qualifies; spread 94¢	untestable

Appendix C Robustness and extra figures

These figures back the regime, replay, latency and capacity claims in the main text.

a Edge stays positive in every venue regime **b** 7 of 7 in-sample months are positive

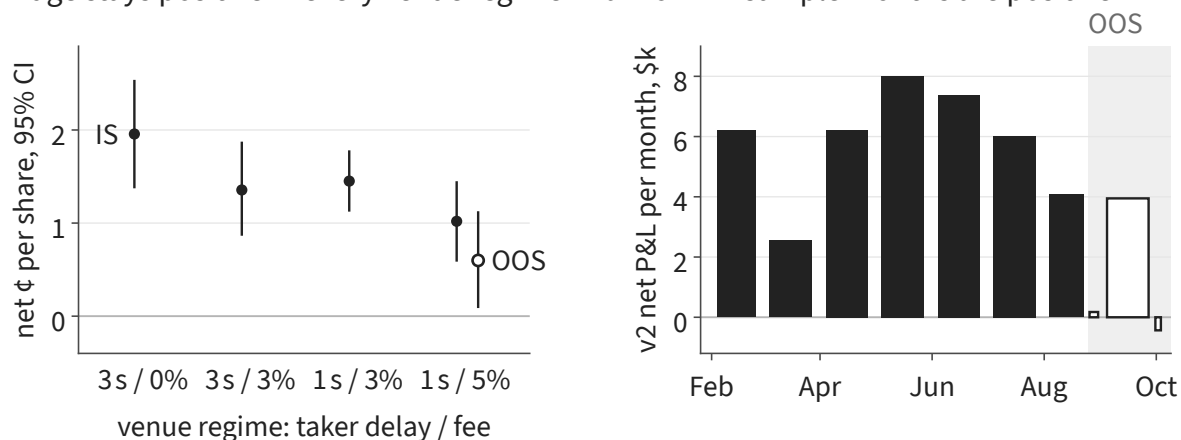
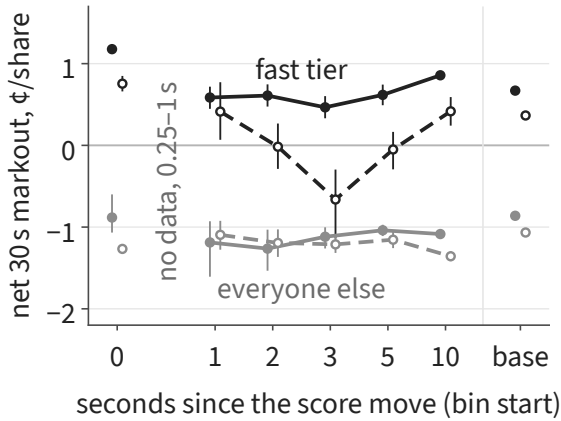


Figure A1. v2's edge shrinks as the venue's protection shrinks, and stays positive in every regime. (a) Net ¢ a share by venue regime (taker hold / fee), 95% CIs; (b) net P&L by month, at the fast tier's own prices. Source: results/risk/risk_stats.json.

a The fast tier stays ahead at every horizon



b Trading every point loses at every delay

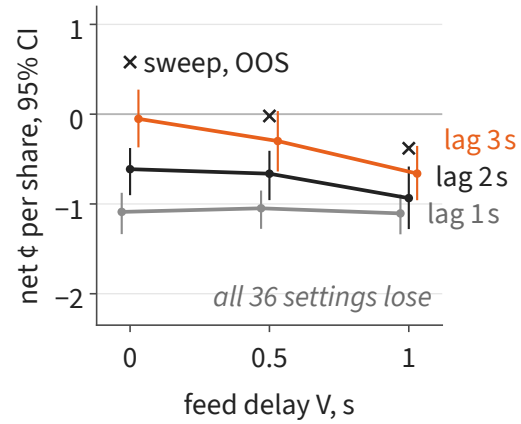
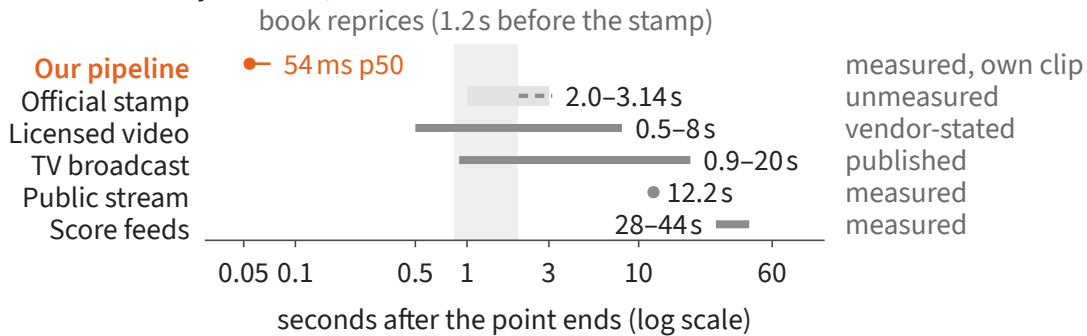


Figure A2. *The fast tier leads at every horizon; trading every point on real books loses.* (a) Net 30 s markout by seconds since the score move; filled: IS, hollow: OOS. (b) Replay of 9 matches recorded live on 2026-10-03 against their real order books, every point called; net ¢ a share marked at +30 s with match-clustered 95% CIs; ×: the sweep's OOS cell. One day, assumed feed latency, paper only. Source: results/{decay,replay}.

a Our order is ready in 54 ms; feeds take seconds



b Live engine: 4 of 41 misses called early, none wrong

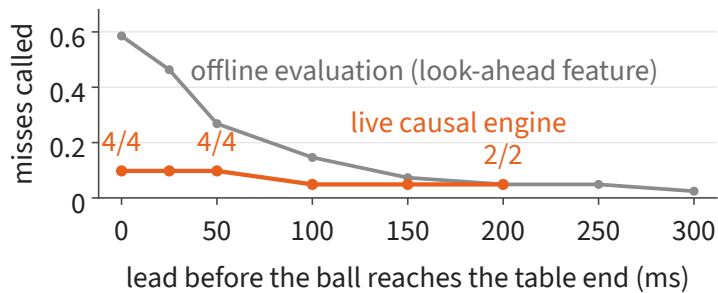
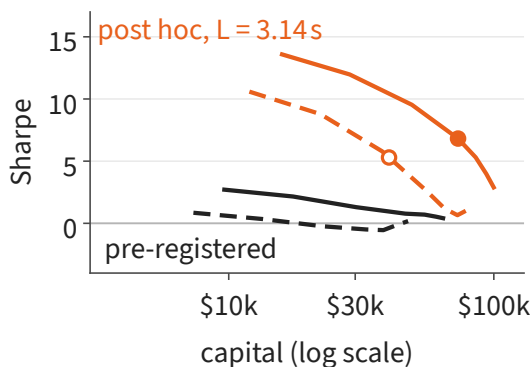


Figure A3. *Who sees the point first.* (a) Delay after the point ends (log scale): our pipeline (dot p50, whisker p99, own clip over WebRTC, laptop); the official stamp, dashed from the pre-registered 2.0s to the post hoc 3.14s over the cited 1–3s band (Regen Sports, 2026; Stats Perform, nd); vendor-stated licensed video (Stats Perform, nd; NEXT.io, nd; Sportradar, nd; Aply, nd); published stream delays (ISPreview, 2021; GL Systemhaus, 2018; The Desk, 2026); measured public stream and score feeds. Shaded: when the book reprices. (b) Early out-calls on 41 held-out table-tennis misses: the live causal engine (orange), which we can deploy, and the offline evaluation (grey), which used a look-ahead feature. Source: results/{e2e,webrtc,tracking,engine}, research/v2/latency.

a Sharpe halves by \$40k–\$73k (post hoc)



b The cheapest data stack needs \$14k–\$16k

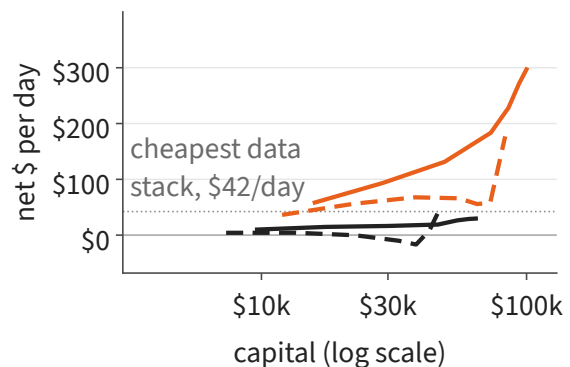


Figure A4. *How much capital the CV strategy can use at a 1s feed.* (a) Sharpe and (b) net \$ a day against capital (net cap 50 to 5,000 shares, \$250 orders, 10 matches a day, half the stale depth); circles: where the post hoc Sharpe halves; dotted: the cheapest data stack. Solid: IS; dashed: burned OOS. Simulated, assumed feed latency (licensed feed not purchased); parameters measured. Source: results/capacity.

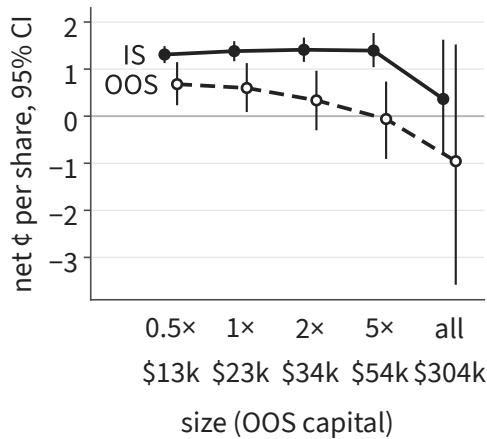
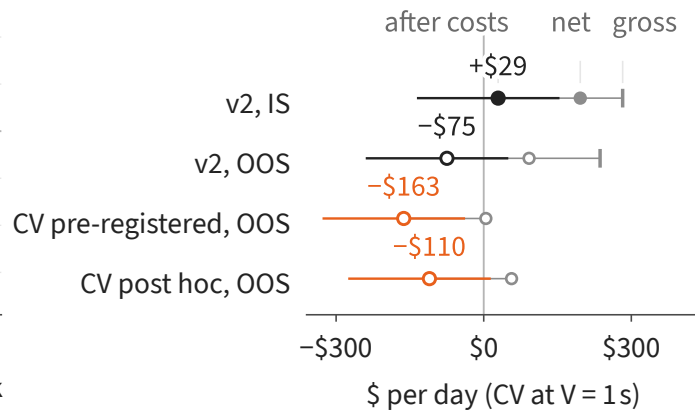
a v2's OOS edge fades as size grows**b** After fixed costs only v2 IS stays positive

Figure A5. *v2's capacity is small and fixed costs bind.* (a) Net edge by size (caps scaled; optimistic at large size, since we model no price impact); filled: IS, hollow: OOS. (b) \$ a day gross, after fees and after central fixed costs (bar: low to high); CV rows at the 1s base case, burned OOS, pre-registered first; assumed feed latency (licensed feed not purchased); parameters measured. Source: results/financials.

Appendix D Records

We keep the full logs in the repo instead of printing them here.

Pre-registration. H1–H4: HYPOTHESIS.md, committed at 7232986 (2026-10-03 09:50 UTC) before any result. H6, v2 and the forward test: HYPOTHESIS_V2.md (amendments A1–A5). Table tennis: HYPOTHESIS_TT.md. The CV simulation's research/v2/tier0/PREREG.md says it was written at 17:25 UTC; it was committed with its results at 19:24 UTC (a5769c7), and the peek log shows the OOS grid first ran at 17:29 UTC. git log on these files lists 19 commits.

Deviations. DEVIATIONS.md and research/v2/*/DEVIATIONS*.md give every rule change and its reason: 2 changes after a look for v2 and 13 for the CV simulation (T1–T3, V1–V10).

Reads of held-out data. results/oos_peeks.log has 77 lines, one per read: 7 blind first runs, 16 non-blind reads, 5 descriptive checks, 25 audits or re-runs, 22 live-data or replay reads and 2 others; 2 were written after their run and say so.

Variants. 4,219 in all (results/paper/variants.json): H1–H6 pre-registered and H5/H6 rules (44); v2 lenses (exit, sizing, selection, cross-market, latency, Kalshi) (3,342); v2-safe grid (24); tier-0 pre-registered scenarios (432); tier-0 v3 in-sample grid (360); maker v1 evaluated variants (8); cost-stress cases (4); table-tennis tests TT1–TT5 (5). Sensitivities that chose nothing: 944 latency-sweep cells and 36 replay cells, plus 4 rally-gate values (Section 6). Two presentation choices were post hoc: the 3.14s headline reading and the 1s base case.

Grids and sources. Latency sweep: results/tier0/latency_sweep.json; replay: results/replay/selective/selective.json; fixed costs: results/financials/financials.json; latency sources: research/v2/latency/results.json; every number with its source: results/paper/numbers.json.

Forward test and live session. Blind forward test: pre-registered but not run within the hackathon window (HYPOTHESIS_V2.md A5). The live paper session was stopped by a team decision and is not used (research/v2/maker/DEVIATIONS_LIVE.md, L15).